FOUR TESTING CRITERIA FOR IND-QUASI-AFFINE MORPHISMS AND ZARISKI LOCALITY ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19537-19557 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19559-19582, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Extends the lane's fpqc-local companion card L16080 (ind-quasi-affine-fpqc-local): this is the Zariski-side structural dictionary used by that descent proof; the in-particular locality sentence is retained.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of ind-quasi-affine morphism (chapter definition section), Morphisms, lemmas characterize-quasi-affine and quasi-affine-permanence; Schemes, lemmas quasi-compact-permanence and compose-after-separated.

Source reference. Original lemma 01SM: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19582–19582. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 054G: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19582–19582. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01KV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19582–19582. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03GI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19582–19582. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *For a morphism of schemes $f : X \to Y$, the following are equivalent:*

(1) *f is ind-quasi-affine,*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(2) *for every affine open subscheme $V \subset Y$ and every quasi-compact open subscheme $U \subset f^{-1}(V)$, the induced morphism $U \to V$ is quasi-affine.*
(3) *for some cover $\{V_j\}_{j \in J}$ of Y by quasi-compact and quasi-separated open subschemes $V_j \subset Y$, every $j \in J$, and every quasi-compact open subscheme $U \subset f^{-1}(V_j)$, the induced morphism $U \to V_j$ is quasi-affine.*
(4) *for every quasi-compact and quasi-separated open subscheme $V \subset Y$ and every quasi-compact open subscheme $U \subset f^{-1}(V)$, the induced morphism $U \to V$ is quasi-affine.*

In particular, the property of being an ind-quasi-affine morphism is Zariski local on the base.

Proof. The equivalence (1) $\Leftrightarrow$ (2) follows from the definitions and Morphisms, Lemma 01SM. For (2) $\Rightarrow$ (4), let U and V be as in (4). By Schemes, Lemma 03GI, the induced morphism $U \to V$ is quasi-compact. Thus, for every affine open $V' \subset V$, the fiber product $V' \times_V U$ is quasi-compact, so, by (2), the induced map $V' \times_V U \to V'$ is quasi-affine. Thus, $U \to V$ is also quasi-affine by Morphisms, Lemma 01SM. This argument also gives (3) $\Rightarrow$ (4): indeed, keeping the same notation, those affine opens $V' \subset V$ that lie in one of the V_j cover V, so one needs to argue that the quasi-compact map $V' \times_V U \to V'$ is quasi-affine. However, by (3), the composition $V' \times_V U \to V' \to V_j$ is quasi-affine and, by Schemes, Lemma 01KV, the map $V' \to V_j$ is quasi-separated. Thus, $V' \times_V U \to V'$ is quasi-affine by Morphisms, Lemma 054G. The final implications (4) $\Rightarrow$ (2) and (4) $\Rightarrow$ (3) are evident. $\square$